

Car Price Prediction System

1. Introduction

In the rapidly growing automobile industry, determining the correct resale value of a car is a complex and time-consuming task. Car prices are influenced by multiple factors such as brand reputation, model type, fuel category, transmission, manufacturing year, mileage, and usage history. Traditional pricing methods often rely on human judgment, which can be inaccurate and inconsistent.

The **Car Price Prediction System** is designed to overcome these challenges by applying **Machine Learning techniques** to analyze historical car data and predict accurate resale prices. By learning patterns from past sales, the system provides reliable, data-driven price predictions that help buyers and sellers make informed decisions. This project forms the Machine Learning foundation for a full-stack application developed using the **MERN stack and Angular frontend**.

2. Problem Statement

Car buyers and sellers often face difficulties in estimating a fair market price for used cars due to:

- Lack of market knowledge
- Price variation across regions
- Dependence on dealers or third-party platforms
- Manual estimation errors

Hence, there is a need for an **automated system** that predicts car prices accurately using historical data and machine learning models.

3. Objectives of Data Collection & Preprocessing (ML Part-A)

The main objectives of this phase are:

- To collect a reliable and relevant car dataset
 - To clean and preprocess raw data
 - To convert categorical features into numerical form
 - To engineer meaningful features for better prediction
 - To prepare a high-quality dataset for model training
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4. Dataset Description

The dataset used in this project was collected from **Kaggle / CSV sources** containing historical car sales data.

Key Attributes:

- Car Brand
- Car Model
- Year of Manufacture
- Fuel Type
- Transmission
- Mileage

- Kilometers Driven
 - Selling Price (Target Variable)
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5. Data Analysis & Preprocessing

5.1 Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand:

- Data distribution
- Feature importance
- Presence of outliers
- Correlation between variables

This step helped in identifying issues such as missing values, duplicate records, and inconsistent entries.

5.2 Data Cleaning

The following cleaning steps were performed:

- Removal of duplicate records
- Handling missing values using mean, median, or mode
- Removal of irrelevant or noisy columns
- Standardization of data formats

These steps improved data consistency and reliability.

5.3 Encoding Categorical Features

Since machine learning models work with numerical data, categorical variables such as:

- Brand
- Model
- Fuel Type
- Transmission

were converted into numerical format using **label encoding / one-hot encoding** techniques.

5.4 Feature Engineering

To improve prediction accuracy, new features were created:

- **Car Age** = Current Year – Manufacturing Year
- Removal of redundant features
- Normalization of numerical values

Feature engineering helped the model better understand real-world patterns.

6. Final Preprocessed Dataset

After completing all preprocessing steps:

- The dataset became clean and structured
 - All features were machine-learning ready
 - Noise and inconsistencies were removed
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7. Deliverables (Member-1)

`cleaned_data.csv`

- Final processed dataset
- Ready for ML model training
- Free from missing values and duplicates

Data Preprocessing Notebook

- Step-by-step preprocessing implementation
 - Code for cleaning, encoding, and feature engineering
 - Acts as documentation and reproducibility reference
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8. Importance of Data Preprocessing

Proper data preprocessing:

- Improves model accuracy
- Reduces overfitting
- Enhances prediction reliability
- Saves computational resources

This phase directly impacts the performance of all machine learning models used in later stages.

9. Conclusion

The Data Collection and Preprocessing phase is the backbone of the **Car Price Prediction System**. By carefully cleaning, analyzing, encoding, and engineering features, raw car sales data was transformed into a high-quality dataset suitable for machine learning. This well-prepared dataset enables effective model training and accurate price prediction in subsequent phases of the project. The successful completion of this phase ensures a strong foundation for model building, API integration, and full-stack deployment.